RYERSON POLYTECHNIC UNIVERSITY

DEPARTMENT OF MATHEMATICS, PHYSICS AND COMPUTER SCIENCE

FINAL EXAM

MTH 240 – Calculus II

Last Name (Print):_____. First Name:_____. Student Number: _____

Signature:_____.

Professor (circle one)

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Duration: 3 hours

1. This is a closed-book exam. Notes, calculators and other aids are not permitted. Verify that your exam has pages 2-8.

2. **Section A** is multiple-choice. Check that the "bubble sheet" at your desk has **your name and student number on it**. The first question asks you to identify the colour of your test paper. Fill this in at the **beginning** of the exam. Make sure to circle your answers in this booklet first, then copy them to the bubble sheet **carefully**. There are no part-marks in the multiple-choice section and **only** the bubble sheet will be marked. The correct response gets full marks, an incorrect response or no response gets no marks. *Read all responses to the multiple choice questions before making a selection.*

3. **Section B** is full-answer.

   (a) Unless otherwise instructed, **make sure you include all significant steps in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution.** Put a box around your final **answer.**

   (b) Write your solutions in the space provided. If you need more space, use the back of the page. Indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there. Marks (out of 100) are shown in brackets.

4. Do not separate the test sheets.

5. At the end of the Exam place the bubble sheet in the booklet and hand them in together.

For Instructor's use only.

Page	Mark
2 - 3	
4	
5	
6	
7	
8	
Total	

Part A. Multiple-Choice [Questions 2-15, 3 marks each] (Choose one answer in each case.)

1. What colour is this test paper? (Mark this on the bubble sheet immediately.)
   A) White. B) Tan
2. Consider the following limits:

   i) $\lim_{x\to 0+} \frac{\sin(x)}{\csc(x)}$ ii) $\lim_{x\to 0+} (1+x)^x$ iii) $\lim_{x\to 0+} (\csc(x) - \cot(x))$

   Which of the above is an indeterminate form?

   A) Only i). B) Only ii). C) Only iii). D) Only i) and ii). E) Only ii) and iii).
3. Consider the following integrals:

   i) $\int_0^\infty e^{-x}\,dx$ ii) $\int_0^4 \frac{x}{x^2+2}dx$ iii) $\int_0^\pi \ln(x)dx$ iv) $\int_0^2 \frac{e^x}{x-1}dx$

   Of these integrals, which one(s) is(are) improper?

   A) All. B) Only i). C) Only i) and iv). D) Only i), ii) and iii). E) Only i), iii) and iv).

4. Consider the following statements.

   i) If a function f is continuous on $[0,\infty)$ and $\lim_{x\to\infty} f(x) = 0$ then $\int_0^\infty f(x)dx$ converges.

   ii) If a function f is continuous on $[0,\infty)$ and $\int_0^\infty f(x)dx$ diverges then $\lim_{x\to\infty} f(x) \neq 0$.

   iii) If f' is continuous on $[0,\infty)$ and $\lim_{x\to\infty} f(x) = 0$ then $\int_0^\infty f'(x)dx = -f(0)$.

   Choose the correct option below.

   A) Only i) is correct. B) Only ii) is correct. C) Only iii) is correct.

   D) i), ii), and iii) are all correct. E) Only i) and iii) are correct.

5. An integral of the form $\int f(\sqrt{a^2-u^2})\,du$ may be evaluated using a trigonometric substitution. Choose the correct substitution.

   A) Use $u = a\sin(\theta)$ to give $\int(f(a\cos(\theta))a\sin(\theta)\,d\theta$ B) Use $u = a\sin(\theta)$ to give $\int(f(a\cos(\theta))a\cos(\theta)\,d\theta$

   C) Use $u = a\sec(\theta)$ to give $\int(f(a\tan(\theta))a\sec(\theta)\,d\theta$ D) Use $u = a\tan(\theta)$ to give $\int(f(a\sec(\theta))a\sec^2(\theta)\,d\theta$

   E) None of the above

6. The *sequence* $\{a_n\}$ where $a_n = n\sin(2/n)$:

   (A) converges to 0 (B) converges to 1 (C) converges to 2 (D) diverges (E) None of the above.

7. The *infinite series* $\sum_{n=0}^{\infty} r^n$:

   (A) Converges only for $0 \le r < 1$ (B) Converges only for $-1 < r \le 0$

   (C) Converges only for $-1 \le r \le 0$ (D) Diverges only for $|r| > 1$ (E) Converges only for $|r| < 1$.

8. The **Alternating Series test** may be used:

   i) **To establish** the divergence of $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{2n^2-3n+4}$

   ii) **To establish** the **absolute** convergence of $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$

   iii) **To establish** the convergence of $\sum_{n=1}^{\infty} (-1)^n \frac{n^2-n}{n^2+1}$

   Choose the correct option below.

   A) Only i) is correct. B) Only ii) is correct. C) Only iii) is correct.

   D) i), ii), and iii) are all correct. E) None of the above

9. Consider the following statements:

   i) If $\sum_{n=0}^{\infty} a_n x^n$ converges for $x = 2$ then it also converges for $x = -2$.

   ii) If $\sum_{n=0}^{\infty} a_n$ converges then so does $\sum_{n=0}^{\infty} |a_n|$.

   Choose the correct option below.

   A) Only i) is correct. B) Only ii) is correct. C) Both i) and ii) are correct.

   D) Both i) and ii) are false.

   E) None of the above

10. The infinite series $\sum_{n=0}^{\infty} r^{3n}$ for $|r| < 1$

    A) Converges to $\frac{r^2}{1-r}$. B) Converges to $\frac{r^3}{1-r^3}$. C) Converges to $\frac{1}{1-r^3}$.

    D) Diverges. E) None of the above.

11. The point $(x, y) = (x_0, y_0)$ is a critical point of the function $f(x, y)$. Furthermore, $f_{xx}(x_0, y_0) = -3$, $f_{yy}(x_0, y_0) = -8$, $f_{xy}(x_0, y_0) = -5$, and the partial derivatives are all continuous. Then:

    A) $f(x_0, y_0)$ is a local minimum of f. B) $f(x_0, y_0)$ is a local maximum of f.

    C) $f(x_0, y_0)$ is a saddle point of f. D) The second derivative test fails.

    E) None of the above.

12. A surface is specified by $z = e^{-2x}\cos(3y)$. At the point (0,0,1) the slope of the surface in the x-direction is:

    A) -3 B) -2 C) -1 D) 0 E) None of the above.

13. Consider the functions: $f(x, y) = x^3 - 4xy^2 + y^3$, $g(x, y) = \frac{xy}{\sqrt{x^2+y^2}}$ and $h(x, y) = \frac{x^2-y}{x-1}$

    Choose the correct option below.

    A) f is homogeneous of degree 1, g is homogeneous of degree 2, and h is homogeneous of degree 0.

    B) f is homogeneous of degree 3, g is homogeneous of degree 0, and h is homogeneous of degree 1.

    C) f is homogeneous of degree 0, g is homogeneous of degree 3/2, and h is homogeneous of degree 1.

    D) f is homogeneous of degree 3, g is homogeneous of degree 1, and h is not homogeneous.

    E) None of the above

14. Consider the following statements:

    i) The differential equation $2xy\,dx + (y^2 - x^2)\,dy = 0$ is exact.

    ii) The differential equation $y' + xy^2 = \cos(y)$ is linear.

    iii) The separable differential equation $f(x)\,dx + g(y)\,dy = 0$ is also exact.

    Choose the correct option below.

    A) Only i) is correct. B) Only ii) is correct. C) Only iii) is correct.

    D) i), ii), and iii) are all correct. E) Only i) and iii) are correct.

Part B. Full-Answer

15. [4 marks] Solve the differential equation $\dfrac{dy}{dx} = \tan^3(x)\,\sec(x)$

16. [6 marks] **Determine the convergence or divergence** of the following integral. Evaluate the integral if it converges. $\displaystyle\int_0^1 \ln(x)\,dx$

17. [4 marks] Find the power series expansion of $f(x) = 3^x$ about the point $c = 0$.

18. [10 marks] Determine whether each of the following series converges conditionally, converges absolutely or diverges. **State the name of the tests you are using and all the requirements of the tests in each case.**

(a) $S = \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln(\sqrt{n})}$

(b) $S = \sum_{n=1}^{\infty} \frac{\arctan(n)}{1+n^2}$ [You may **assume** $\{ \frac{\arctan(n)}{1+n^2} \}$ is decreasing.]

19. [3 marks] Let $w = f(x, y)$, $x = g(s, t)$ and $y = h(s, t)$. **Use the appropriate chain rule for multivariable functions** to write formulas for $\partial w/\partial s$ and $\partial w/\partial t$.

20. [4 marks]Use the total differential of $z = 9 - x^2 - y^2$ to find the approximate change in z if x changes from 1 to 1.05 and y changes from 2 to 1.9. Simplify your answer.

21. [6 marks] Find and identify all local maxima, minima and saddle points of the surface defined by $h(x, y) = x^2 - 6xy + 10y^2 - 4y + 4$.

22. [3 marks] A 400 litre tank initially contains 50 litres of pure water. Brine with a concentration of 3 grams of salt per litre of water runs into the tank at the rate of 7 litres per minute and is uniformly stirred into the solution. Meanwhile, the mixture runs out of the tank at the constant rate of 5 litres per minute. Let $y(t)$ denote the amount of salt in the tank at time t. Write down a differential equation for $y(t)$ and the appropriate initial condition (**DO NOT SOLVE THE DIFFERENTIAL EQUATION**).

23. [6 marks] Find the general solution of $\dfrac{y}{x-1}\,dx + [\ln(x-1) + 2y]\,dy = 0$

24. [6 marks] Find the general solution of the differential equation $y' = \dfrac{3x+2y}{x}$.

25. [3 marks] Find the **differential equation** for the orthogonal trajectories of the family of curves given by $y^2 = cx^3$. **DO NOT SOLVE THE DIFFERENTIAL EQUATION**

26. [6 marks] Solve the differential equation $y' + \frac{y}{x} = e^{-x}$

27. [5 marks] **Bonus Problem: Do not attempt this unless you have attempted to complete the earlier questions.** Consider the ordinary differential equation $y' + 3x^2y = x^2y^3$. The equation is neither separable, homogeneous, exact or linear.

(a) Show that the differential equation *is* linear in the variable $z = y^{-2}$.

(b) Find the general solution of the differential equation for z.

(c) Find the general solution for y.